# Dat Nguyen

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## Education

**University of California, San Diego** La Jolla, CA
*B.S. in Data Science and B.A. in Economics* *Sept. 2024 – Jun. 2028*

- **Relevant Coursework:** Principles of Data Science, Data Structures and Algorithms, Linear Algebra, Calculus I–III, Principles of Economics, Macroeconomics, Programming for Data Science

## Projects

**Modeling Housing Affordability Trends in San Jose** Jan 2026 – Present
*Python, pandas, NumPy, Matplotlib, Seaborn, Jupyter Notebooks*

- Analyzed **housing affordability trends** in the San Jose MSA using **130+ monthly observations** of home prices, rents, and household income
- Cleaned and reshaped **Zillow** and **U.S. Census** datasets into a unified **time-series dataset**
- Performed exploratory **time-series analysis** and visualizations to evaluate long-term housing cost trends using **ARIMA** and **SARIMA**

**Smart Temperature-Controlled Energy Dashboard** Jan 2026 – Mar 2026
*ESP32, Arduino IDE, Python, pandas, Supabase, Streamlit*

- Collected **real-time temperature and power data** by programming an **ESP32 microcontroller** to transmit sensor readings to **Supabase**
- Built an **IoT monitoring system** that streams sensor data from ESP32 devices for cloud storage and analysis
- Developed an interactive **Streamlit dashboard** to visualize power usage (W), cumulative energy consumption, and projected device impact
- Estimated **energy cost savings** by analyzing temperature-threshold control strategies using a fan as a case-study device

**Statistical Analysis of Food Delivery Platform Performance** Jan 2026 – Mar 2026
*Python, pandas, Seaborn, Matplotlib, Statsmodel, Scipy*

- Analyzed **21,000+ food delivery transactions** from Grubhub (8k), DoorDash (11k), and FoodHub (2k) to investigate drivers of customer satisfaction
- Identified relationships between delivery time, preparation time, cost, and ratings using **exploratory data analysis (EDA)**
- Compared platform performance by applying statistical methods including **t-tests**, **Mann–Whitney U tests**, and **Spearman correlations**

**FRIENDS: Data-Driven Analysis for Enhanced Engagement** Jan 2025 – Mar 2025
*Python, pandas, Jupyter Notebooks*

- Modeled relationships between character roles, episode ratings, and audience engagement using **statistical and Bayesian models**
- Identified factors influencing viewership by applying **bootstrapping** and **hypothesis testing**
- Communicated audience insights through data visualizations and exploratory analysis

## Experience

**CALPIRG Students – UC San Diego Chapter** 2024 – Present
*Volunteer Advocate* *La Jolla, CA*

- Collaborated with student teams to support campus outreach and advocacy initiatives
- Assisted in organizing campaigns focused on environmental and community engagement topics
- Developed communication and leadership skills through public-facing events

## Technical Skills

**Languages**: Python, Java, HTML/CSS, LaTeX, MATLAB
**Libraries & Frameworks**: pandas, NumPy, Matplotlib, Seaborn, Streamlit, statsmodel, Scikit-Learn, SciPy
**Data & Statistical Methods**: Hypothesis Testing, Bootstrapping, Bayesian Modeling, t-tests, ANOVA, Effect Size Analysis, Data Cleaning, Data Visualization, K Nearest Neighbors, SVM, ARIMA, SARIMA
**Tools & Platforms**: Git, Jupyter Notebooks/Lab, IntelliJ, Overleaf, Arduino IDE, Anaconda, Supabase, Excel, Google Sheets, Canva